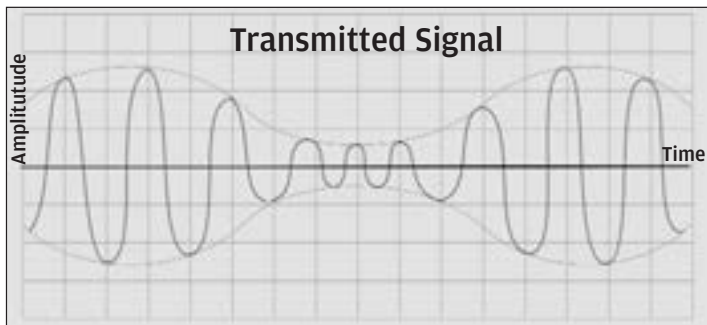


When you speak into a cell phone (or any radio device for that matter), your voice is converted into an electrical signal. This signal, though, cannot be transmitted into the air as it is - it will gradually lose its strength in the face of air currents, dust and other obstacles. It would be a tall order for a piece of equipment to be able to accurately receive all the different frequencies of a human voice by themselves. To get around this, the phone makes the sound waves hitch a ride on a much more powerful, high-frequency wave called a carrier. This process is called *modulation*. The carrier wave travels at the speed of light, so your voice is almost instantaneously carried to the intended receiver.

There are two ways to modulate a voice signal - Amplitude modulation (AM) and Frequency Modulation (FM). In AM, changes in the *amplitude* (intensity or loudness) of the voice signal causes changes in the amplitude of the carrier wave. In FM, the amplitude of the voice signal causes changes in the *frequency* of the carrier wave.



An amplitude modulated wave; the carrier amplitude increases and decreases with the voice amplitude.

Because our voices aren't composed of one frequency alone, modulating them causes the resulting wave to get spread over a band of frequencies - so a typical modulated wave would not just have the carrier frequency, but would be distributed over a range of frequencies. This is called a *frequency band*.

Modulating a sound wave solves several problems. For one